

CSA0609 - DESIGN AND ANALYSIS OF ALGORITHMS

Name: Soujanya.B

Reg No: 192521389

Code: CSA0609

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DAA LAB EXPERIMENT 4

Binary Search with Big-O Analysis

Aim

To implement Binary Search in Python and analyze its time and space complexity using Big-O notation.

Algorithm

1. Start.
2. Take a **sorted array** as input.
3. Set $low = 0$ and $high = n - 1$.
4. Find the middle element.
5. Compare the middle element with the search key.
6. If they are equal, return the position.
7. If the key is smaller, search the left half.
8. If the key is larger, search the right half.
9. Repeat until the element is found or the search range becomes empty.
10. Stop.

Python Source Code

```
def binary_search(arr, key):
```

```
    low = 0
```

```
    high = len(arr) - 1
```

```
    while low <= high:
```

```
        mid = (low + high) // 2
```

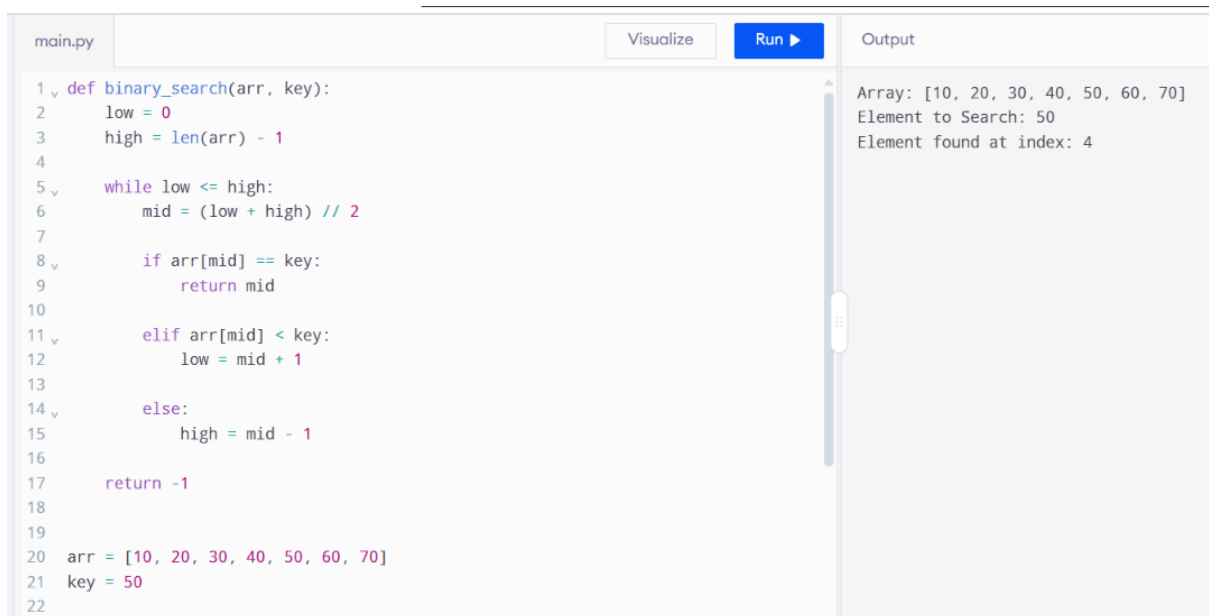
```
        if arr[mid] == key:
```

```
            return mid
```

```
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    return -1

arr = [10, 20, 30, 40, 50, 60, 70]
key = 50
result = binary_search(arr, key)
print("Array:", arr)
print("Element to Search:", key)
if result != -1:
    print("Element found at index:", result)
else:
    print("Element not found")
```

Output



The screenshot shows a Python IDE with a file named 'main.py'. The code defines a 'binary_search' function and uses it to search for the value 50 in the array [10, 20, 30, 40, 50, 60, 70]. The output panel on the right displays the results of the program execution.

```
main.py Visualize Run ▶ Output
```

```
1 def binary_search(arr, key):
2     low = 0
3     high = len(arr) - 1
4
5     while low <= high:
6         mid = (low + high) // 2
7
8         if arr[mid] == key:
9             return mid
10
11        elif arr[mid] < key:
12            low = mid + 1
13
14        else:
15            high = mid - 1
16
17    return -1
18
19
20 arr = [10, 20, 30, 40, 50, 60, 70]
21 key = 50
22
```

Array: [10, 20, 30, 40, 50, 60, 70]
Element to Search: 50
Element found at index: 4

Result

Thus, the Binary Search algorithm was successfully implemented in Python, and its time and space complexities were analyzed using Big-O notation.